Recitation 3

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Rutgers University

Question 1

Let $A, B \subseteq (0, \infty)$ be non-empty and bounded above. Set

$$\alpha = \sup A, \qquad \beta = \sup B,$$

and define

$$C = \{xy : x \in A, y \in B\}$$

By definition of supremum,

$$\alpha \geq x : \forall x \in A$$

and

$$\beta \geq y : \forall y \in B.$$

Then, since all x and y are positive,

$$\alpha\beta \geq xy : \forall x \in A, \forall y \in B.$$

That is, $\alpha\beta$ is an upper bound on C.

Suppose c is an arbitrary upper bound on C. Then

$$c \geq xy$$

for every $x \in A$ and $y \in B$. So

$$\frac{c}{y} \geq x.$$

Then $\dfrac{c}{y}$ is an upper bound on A. Since $\alpha = \sup A$,

$$\frac{c}{y} \geq \alpha,$$

and

$$c \geq \alpha y.$$

Then

$$\frac{c}{\alpha} \geq y$$

That is, $\dfrac{c}{\alpha}$ is an upper bound on B. Since $\beta = \sup B$,

$$\frac{c}{\alpha} \geq \beta,$$

and

$$c \geq \alpha\beta.$$

Therefore, $\alpha\beta$ is also the least upper bound on C or

$$\alpha\beta = \sup C.$$

Question 2

(a) Let S be an infinite set. We construct distinct elements

$$a_1, a_2, a_3, \ldots \in S$$

inductively. Since S is infinite, it is also non-empty. Thus, we can select $a_1 \in S$ as the first element.

Suppose that $a_1, a_2, ..., a_n$ have been chosen for $n \in \mathbb{N}$. Then

$$\{a_1, a_2, ..., a_n\}$$

is still a finite set. Since S is infinite,

$$S \setminus \{a_1, a_2, ..., a_n\} \neq \emptyset$$

So we can choose $a_{n+1} \in S \setminus \{a_1, a_2, ..., a_n\}$. Therefore, by induction, we can choose countably infinitely many distinct elements

$$A = \{a_1, a_2, a_3, \ldots\} \subseteq S.$$

Define this function $f : \mathbb{N} \to A$ as

$$f(n) = a_n.$$

Suppose $f(m) = f(n)$. Then

$$a_n = a_m.$$

Since we define a_i such that each element is distinct, it follows that $m = n$. Thus f is injective.

By definition of a_i, there is some $n \in \mathbb{N}$ for each distinct element a_i. So f is surjective.

Therefore, f is bijective, and

$$\mathbb{N} \sim A \subseteq S.$$

(b) Let $A = \{a_1, a_2, a_3, \ldots\} \subseteq S$ be the countably infinite set constructed in part (a). Define $F : S \to S \setminus \{a_1\}$ as

$$F(x) = \begin{cases} a_{n+1}, & x = a_n \text{ for some } n \in \mathbb{N}, \\ x, & x \notin A. \end{cases}$$

Suppose $F(x) = F(y)$.

If $x, y \in A$, then

$$x = a_m \qquad \text{and} \qquad y = a_n$$

for some $m, n \in \mathbb{N}$. So $F(x) = F(y)$ implies that

$$a_{m+1} = a_{n+1}.$$

Since elements of A are distinct, it follows that

$$m = n.$$

So

$$x = y.$$

If $x, y \notin A$, then $F(x) = x$ and $F(y) = y$. So $F(y) = F(x)$ implies $x = y$.

What about if $x \in A$ and $y \notin A$, or the other way around? Suppose, for contradiction, that $F(x) = F(y)$. Since $x \in A$, it follows that $x = a_i$ for some $i \in \mathbb{N}$. Then $F(x) = a_{i+1}$, but $F(y) = y \notin A$. So $F(x) \neq F(y)$, contradicting our original assumption. Hence this case is impossible.

Thus, F is injective.

Let

$$y \in S \setminus \{a_1\}.$$

If $y \in A$, then $y = a_n$ for some $n \geq 2$. Then

$$F(a_{n-1}) = a_n = y.$$

If $y \notin A$, then

$$F(y) = y.$$

Thus F is surjective.

Therefore F is a bijection, so

$$S \sim S \setminus \{a_1\}.$$

Hence, every infinite set S contains a proper subset that is equivalent to S.

(c) Let $A \subseteq B$, where A is countable and B is uncountable. Suppose, for contradiction, that $B \setminus A$ is countable. Since A is countable, then the union

$$B = A \cup (B \setminus A)$$

must be countable as well. But this contradicts the assumption that B is uncountable. Hence $B \setminus A$ is uncountable.

Since $B \setminus A$ is infinite, by part (a), we can choose distinct elements

$$d_1, d_2, d_3, \ldots \in B \setminus A.$$

Define $h : B \to B \setminus A$ as

$$h(x) = \begin{cases} d_{2n-1}, & x = a_n, \\ d_{2n}, & x = d_n, \\ x, & x \notin A \cup \{d_1, d_2, \ldots\}. \end{cases}$$

Suppose $h(x) = h(y)$. If $x, y \in A$, then

$$d_{2n-1} = d_{2m-1}$$

for some $m, n \in \mathbb{N}$. So $m = n$, and $x = y = a_n$.

Similarly, if $x, y \in \{d_1, d_2, \ldots\}$, then

$$d_{2n} = d_{2m}$$

for some $m, n \in \mathbb{N}$. So $m = n$, and $x = y = d_n$.

If $x, y \notin A \cup \{d_1, d_2, \ldots\}$, then

$$y = x.$$

If x and y belong to different cases, then $h(x) \neq h(y)$ since each of the three cases have disjoint image sets.

Thus, h is injective.

Let $y \in B \setminus A$. Then we have three possibilities. If $y = d_{2n-1}$, then choosing $x = a_n$, we have

$$h(a_n) = d_{2n-1} = y.$$

If $y = d_{2n}$, then choosing $x = d_n$, we have

$$h(d_n) = d_{2n} = y.$$

If $y \notin \{d_1, d_2, \ldots\}$, then choosing $x = y$, we have

$$h(y) = y.$$

Thus h is surjective.

Hence h is a bijection, and so

$$B \sim B \setminus A.$$

If A was instead finite, we simply shift each d_i an extra k elements, where k is the length of A.

e.g.

$$h(x) = \begin{cases} d_i, & x = a_i, \\ d_{n+k}, & x = d_n, \\ x, & x \notin A \cup \{d_1, d_2, \ldots\}. \end{cases}$$

Question 3 Let $\mathcal{F} : \{g : g : \mathbb{R} \to \mathbb{R}\}$ denote the set of all real-valued functions whose domain is $\mathbb{R}$. Assume, for contradiction, that there exists a bijection

$$f : \mathbb{R} \to \mathcal{F}.$$

For each $a \in \mathbb{R}$, let $g_a = f(a)$ be the function corresponding to the real number a.

Now define $h : \mathbb{R} \to \mathbb{R}$ by

$$h(x) = 1 + g_x(x).$$

Notice that $h \in \mathcal{F}$. Let $a \in \mathbb{R}$. Then

$$h(a) = 1 + g_a(a).$$

Hence

$$h(a) \neq g_a(a),$$

and since a is arbitrary,

$$h \neq g_a$$

for every $a \in \mathbb{R}$.

Since $g_a = f(a)$, this means that h is never equal to $f(a)$, and so it is not in the image of f. But $h \in \mathcal{F}$, so this contradicts the assumption that f is surjective on $\mathcal{F}$. Therefore, there is no bijection from $\mathbb{R}$ to $\mathcal{F}$, and so

$$\mathcal{F} \not\sim \mathbb{R}.$$

Question 4 Let $f : (0,1) \to \mathbb{R}$. Assume there is a constant $M > 0$ such that for every finite set of distinct points $x_1, ..., x_k \in (0,1)$, we have

$$\sum_{j=1}^{k} |f(x_j)| \leq M.$$

Let $S = \{x \in (0,1) : f(x) \neq 0\}$. For $n \in \mathbb{N}$, define

$$S_n = \{x \in (0,1) : |f(x)| \geq \frac{1}{n}\}.$$

Suppose, for contradiction, that some S_n is infinite. Then, we can choose infinitely many distinct elements from S_n, say

$$x_1, ..., x_k \in S_n$$

for some $k \in \mathbb{N}$. Since each $x_i \in S_n$,

$$|f(x_i)| \geq \frac{1}{n}.$$

So

$$\sum_{i=1}^{k} |f(x_i)| \geq \sum_{i=1}^{k} \frac{1}{n} = \frac{k}{n}.$$

By Archimedean Property, choose k large enough so that

$$k > Mn,$$

and so

$$\frac{k}{n} > M.$$

Then

$$\sum_{i=1}^{k} |f(x_i)| > M,$$

contradicting our assumption that

$$\sum_{j=1}^{k} |f(x_j)| \leq M.$$

Thus, every S_n must be finite.

Let

$$x \in \bigcup_{n=1}^{\infty} S_n.$$

Then $x \in S_n$ for some $n \in \mathbb{N}$. So $x \in (0,1)$ and

$$|f(x)| \geq \frac{1}{n} > 0.$$

Then $f(x) \neq 0$, so $x \in S$. Hence

$$\bigcup_{n=1}^{\infty} S_n \subseteq S.$$

Conversely, let

$$x \in S.$$

So $x \in (0,1)$ and $f(x) \neq 0$. That is,

$$|f(x)| > 0.$$

By Archimedean Property, there exists $n \in \mathbb{N}$ such that

$$\frac{1}{n} < |f(x)|.$$

Then

$$|f(x)| \geq \frac{1}{n}.$$

So $x \in S_n$ for some $n \in \mathbb{N}$. Furthermore,

$$x \in \bigcup_{n=1}^{\infty} S_n.$$

Hence

$$S \subseteq \bigcup_{n=1}^{\infty} S_n.$$

Therefore

$$S = \bigcup_{n=1}^{\infty} S_n.$$

Since S_n is finite, then S is the countable union of finite sets. Hence S must be countable as well.

Question 5 Let $C \subseteq [0,1]$ be uncountable, and define

$$E = \{a \in (0,1) : C \cap [a,1] \text{ is uncountable}\}.$$

(a) Suppose, for contradiction, that $E = \emptyset$. Then $C \cap [a, 1]$ is countable for every $a \in (0, 1)$. Specifically, for each $n \geq 2$,

$$C \cap [\frac{1}{n}, 1]$$

is countable. Then

$$C \cap (0, 1] = \bigcup_{n=2}^{\infty} (C \cap [\frac{1}{n}, 1]).$$

Notice that the right side is the countable union of countable sets. So $C \cap (0, 1]$ is countable. Adding the finite element 0 means C is countable as well. However, this contradicts our assumption that C is uncountable.

Therefore, E is non-empty.

(b) Let $\alpha = \sup E$. Consider

$$C = [0, 1).$$

Then C is indeed an uncountable subset of $[0, 1]$. Then, for every $a \in (0, 1)$

$$C \cap [a, 1] = [a, 1).$$

Which is also uncountable. Thus, $E = (0, 1)$, and $\alpha = 1$. But

$$C \cap [\alpha, 1] = \emptyset,$$

which is clearly not uncountable.

Therefore, $C \cap [\alpha, 1]$ does not have to be uncountable.

(c) False, take

$$C = \left\{ \frac{1}{n} : n \in \mathbb{N} \right\}.$$

This set is infinite. For any $a \in (0, 1)$, if $\frac{1}{n} \in [a, 1]$, then $\frac{1}{n} \geq a$. So $n \leq \frac{1}{a}$. Then only a finite amount of $n \in \mathbb{N}$ will meet this requirement. Thus $C \cap [a, 1]$ is finite for every $a \in (0, 1)$. Therefore, $E = \emptyset$.